

UNIT 6: DIVERSITY OF LIFE FORMS

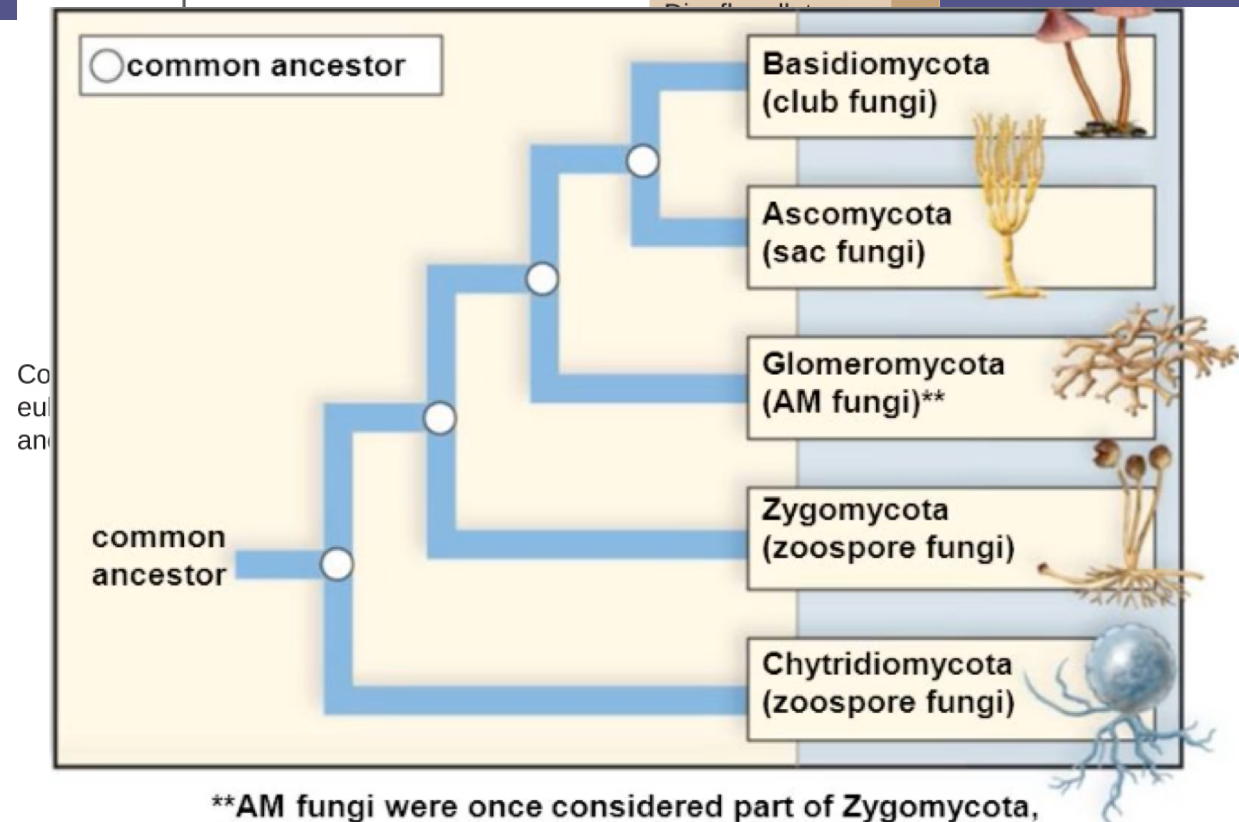
6.4 – FUNGI

Fungi Outline

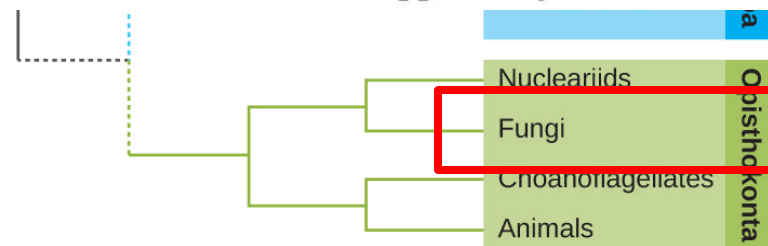
- Evolution
- Structure
- Reproduction
- Diversity
- Symbiotic Relationships of Fungi

Evolution

- Over 80,000 species
- Mostly multicellular, but some unicellular
- Heterotrophic - consume organic matter
 - Saprotrophs – absorb this organic matter
 - Usually dead or decaying matter from surroundings
- Kingdom Fungi
 - 5 Phylum
 - Do not fossilize well, hard to examine fossil record

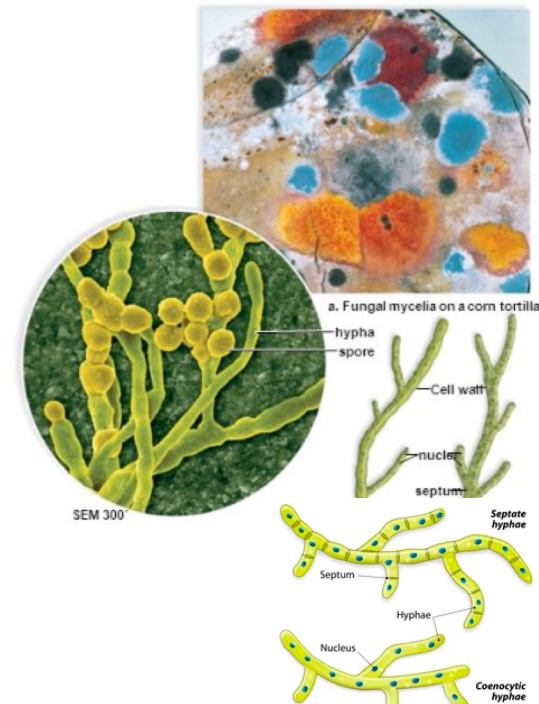


**AM fungi were once considered part of Zygomycota, but molecular data suggest they are a different phylum.



Structure

- Most multicellular
 - ▣ Multicellular usually have a thallus or body structure called **mycelium**
 - Made up of a network of filaments known as **hyphae**
 - Provide high surface area : volume – good for absorption
 - Hyphae can wall off as they grow, becoming **septate**
 - If no cell walls form, **nonseptate**
 - ▣ Lack chloroplasts
 - ▣ Have a cell wall made of chitin instead of cellulose
 - ▣ Most are non-motile
 - Only movement is toward a food source by extension of hyphae



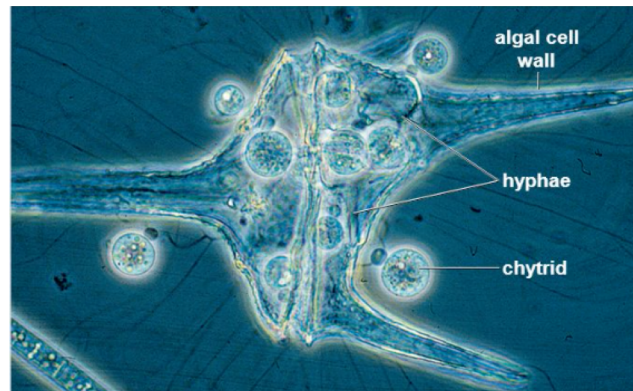
Reproduction

- Use both asexual and sexual reproduction
- Sexual involves 3 stages
 - Haploid Hyphae
 - Dikaryotic Stage
 - Hyphae from two different mating types fuse. Paired haploid nuclei termed dikaryotic.
 - Diploid Zygote
 - Nuclear fusion produces diploid nucleus
 - Haploid spores produced by meiosis
 - Spores germinate directly into haploid hyphae
- Windblown spores allow for spread of fungi since non-motile
- Some species only do asexual through production of windblown spores
- Unicellular yeast reproduce asexually by budding.

Diversity - Chytridiomycota

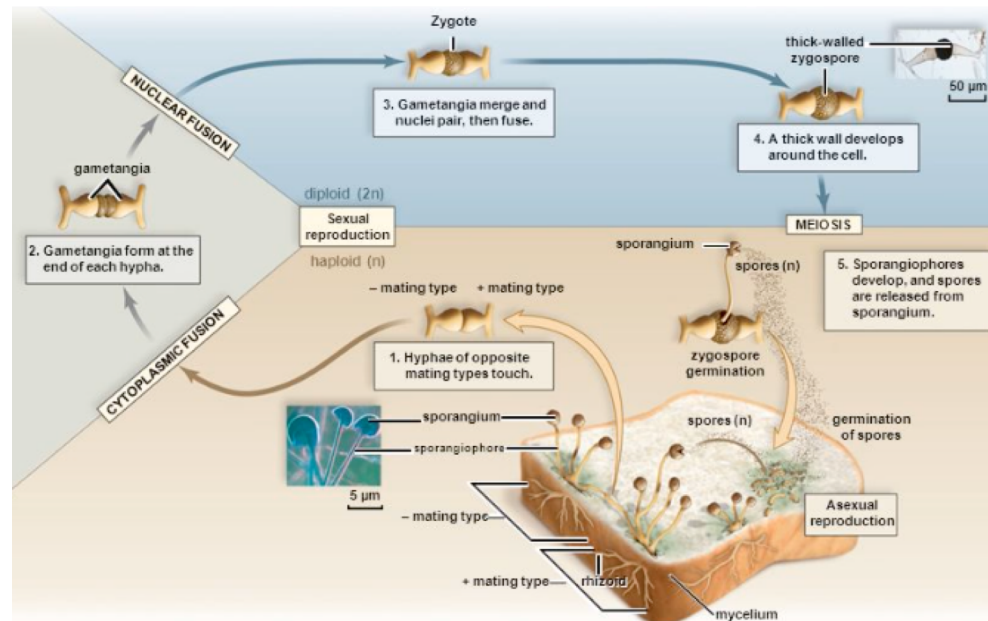
□ Chytrid

- ▣ Aquatic fungi – almost 800 species
- ▣ Unique among fungi
 - Only fungi to still have flagellated cells
 - Help with aquatic lifestyle
 - Produce flagellated gametes and spores
 - Most reproduce asexually through production of zoospore within a single cell, which grows into new chytrid
 - Suggest shared ancestry with choanoflagellate protists



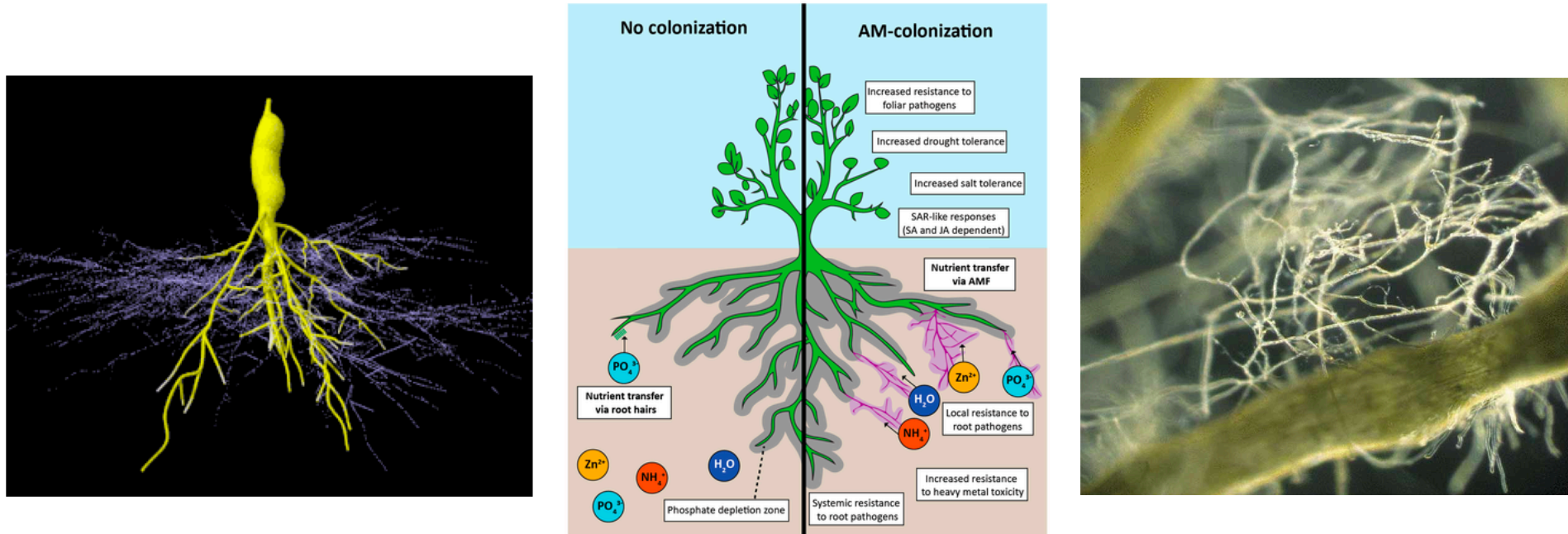
Diversity - Zygomycota

- Mainly saprotrophs – decompose animal and plant matter, common with baked goods
- Black bread mold (*Rhizopus stolonifer*)
 - Both asexual and sexual life cycle
 - Spores settle on food source and develop into 3 specialized hyphae
 - Stolons – grow horizontal on surface of bread
 - Rhizoids – grow into bread as anchors
 - Sporangiophores – Aerial hyphae that produce haploid spores



Diversity – Glomeromycota

- *Arbuscular mycorrhizal (AM) Fungi*
 - ▣ Arbuscules – branching invaginations that fungus make when invading plant roots
 - ▣ Mycorrhizae - mutualistic associations of plants and fungi
 - ▣ Previously grouped as Zygomycota, but molecular data determined it is its own unique genetic group.



Diversity - Ascomycota

- Sac Fungi – 50,000 species
- Most Saprotrophs, most septate hyphae
- Examples
 - Morels and truffles - delicacies
 - Many plant diseases – powdery mildew, ergot of rye...
 - *Aspergillus* & *Candida* cause serious human infection
 - *Talaromyces* (formerly *Penicillium*) is source of penicillin
- Asexual reproduction is the norm
 - Yeast reproduce by budding
 - Small bulge appears on one side of cell, and pinches off
 - Others produce spores called **conidia**, which develop at tip of hyphae
 - These spores can be used for fungal identification

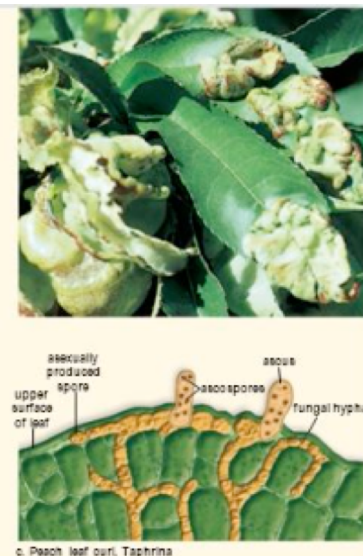
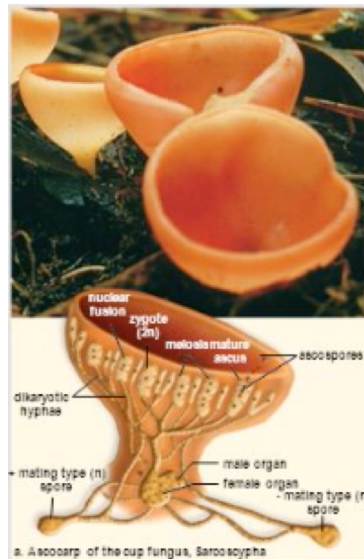


Diversity - Ascomycota

□ Sexual reproduction

■ **Ascus** is a fingerlike sac that develops during sexual

- May be surrounded and protected by sterile hyphae within a fruiting body called an *ascocarp*
 - In cup fungi, *ascocarp* are cup shaped
 - In morels, they are stalked and crowned by a pitted bell-shaped
- Haploid hyphae fuse to make diploid
- Meiosis occurs, followed by mitosis – results in 8 ascospores
 - Spores move by wind, contact female organ, and produce new haploid hyphae



Ascomycota – Benefits and Harms

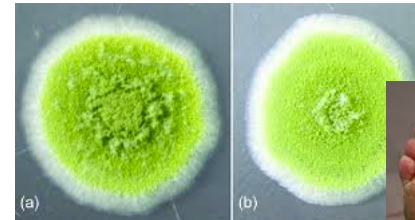
□ Yeast

- Used to make beer and wine, bread leavening
- Genetic experiments – model organism
- *Candida albicans* causes fungal infections – oral thrush



□ Molds

- *Aspergillus* – green molds used to produce soy sauce, and citric and gallic acids
- *Aspergillus flavus* secretes a carcinogenic toxin
- *Trichophyton* – athlete's foot
- *Histoplasma capsulatum* leads to “fungal flu”



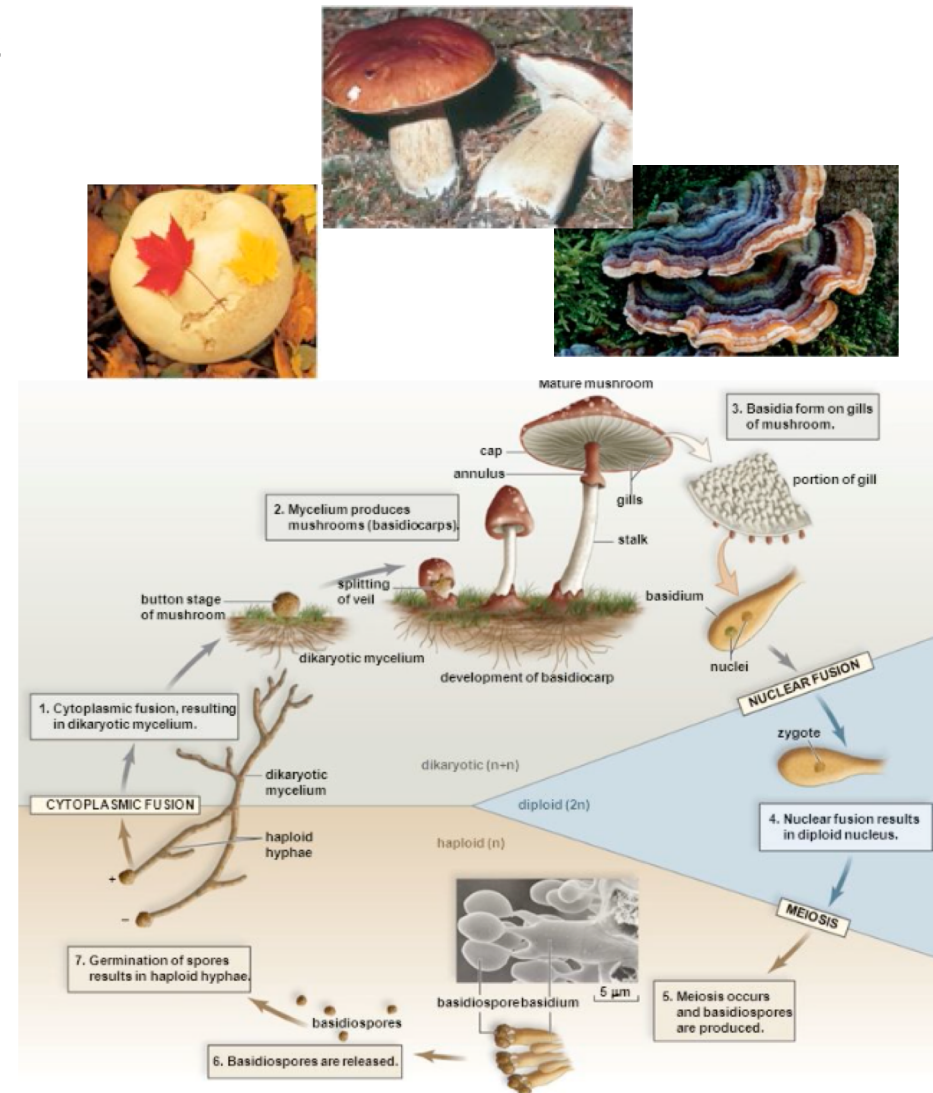
□ Mushrooms

- Yummy
- Certain wild varieties poisonous
 - Red and yellow *Amanitas*
 - Death angel mushroom, *Amanita phalloides*, 90% of fatalities from wild mushrooms
 - Abdominal pain, vomiting, hallucinations
 - Poison inhibits RNA polymerase, leading to kidney and liver damage



Diversity - Basidiomycota

- Club Fungi – 22,000 species
 - Familiar toadstools, mushrooms, bracket fungi, puffballs
 - Some poisonous
 - Some cause plant diseases (smuts and rusts)
 - Mycelium composed of septate hyphae
 - Usually reproduce sexually
 - Haploid hyphae fuse to form dikaryotic mycelium ($n + n$)
 - Dikaryotic mycelium form fruiting bodies – **basidiocarp** with club shaped **basidium**
 - Nuclear fusion followed by meiosis forms **basidiospores**



Symbiotic Relationships of Fungi

□ Lichen

▣ Fungus & Cyanobacterium or green algae

- Fungal hyphae penetrates photosynthetic symbiont
- Nutrients transferred to fungus
 - Possibly mutualistic but fungus more parasitic
 - Photosynthetic symbiont independent
 - Fungal symbiont cannot grow alone

▣ 3 morphological types

- Compact crustose – found on bare rocks or tree bark
- Fruticose – shrublike
- Foliose – leaflike

▣ Can live in extreme conditions

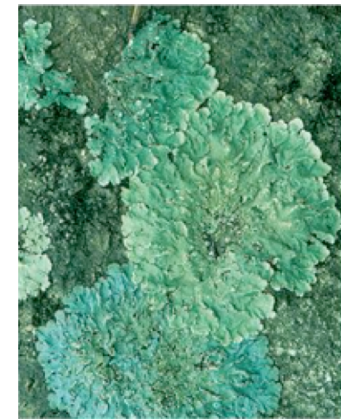
▣ Sensitive indicators of air pollution

□ Mycorrhizae

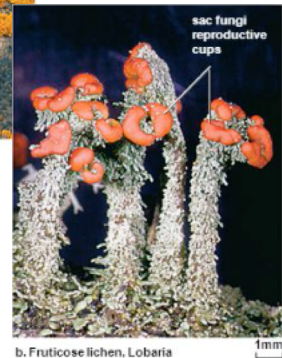
▣ Discussed earlier



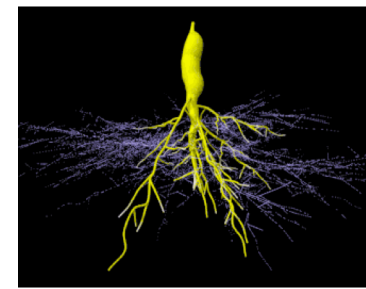
a. Crustose lichen, Xanthoria



c. Foliose lichen, Xanthoparmelia

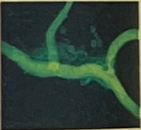


b. Fruticose lichen, Lobaria



Overview

Table 22.1 Features of the Fungi Phyla

Phylum (common name)	Reproduction	Key Features	Examples	
Basidiomycota (club fungi)	basidiospores, sexual	basidiocarp fruiting body (mushroom)	most edible mushrooms, common button mushroom (<i>Agaricus bisporus</i>)	
Ascomycota (sac fungi)	conidiospores, asexual, ascus with spores, sexual	ascocarp fruiting body (mushroom), yeasts, molds	morel mushroom (<i>Morchella</i>), cup fungus (<i>Sarcoscypha</i>)	
Glomeromycota (AM fungi)	Spores, asexual	arbuscules, symbiotic with plants	<i>Glomerales</i>	
Chytridiomycota (chytrid fungi)	zoospores with flagella; most asexual; some alternation of generations	single celled, simplest fungi, aquatic	<i>Chytridiomyces hyalinus</i>	
Zygomycota (zygospore fungi)	zygospores, sexual; sporangiospores, asexual	sporangia, gametangia saprotrophic, feed on animal remains or bakery goods	black bread mold (<i>Rhizopus stolonifer</i>)	